Quadratic Trigonometric Equations

Exam code: 9MA0

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Quadratic trigonometric equations

How do I solve quadratic trigonometric equations?

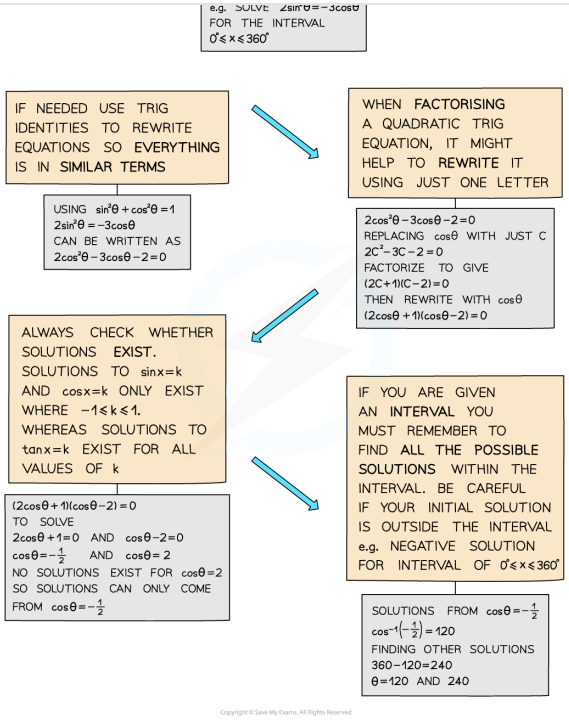
- If an equation involves $\sin^2\theta$ or $\cos^2\theta$ then it is a quadratic trigonometric equation
- These can be solved by factorising and/or using **trigonometric identities** (see **Trigonometry – Simple Identities**)
- As a quadratic can result in two solutions, you will need to consider whether each solution exists and then find all solutions within a given interval for each
- You may be asked to use **degrees or radians** to solve trigonometric equations
  - Make sure your calculator is in the **correct mode**
  - Remember common angles
    - 90° is ½π radians
    - 180° is π radians
    - 270° is 3π/2 radians
    - 360° is 2π radians

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



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Examiner Tips and Tricks

- Sketch the appropriate sin, cos, tan graph to ensure you find ALL solutions **within** the given interval, and be super-careful if you get a negative solution but have a positive interval.
- For example, for an equation, in the interval $0° \leq x \leq 360°$, with solution $\sin x = -¼$ then $\sin^{-1}(-¼) = -14.5$ (to 1d.p.), which is not between 0 and 360 – by sketching the graph you'll be able to spot the two solutions will be 180 + 14.5 and 360 - 14.5.

Worked Example

Giving each solution correct to 1 decimal place.

b) i) Show that $11 \sin x = 5 \cos^2 x + 7$ can be written in the form $5 \sin^2 x + 11 \sin x - 12 = 0$.

ii) Find all the solutions of the equation $11 \sin x = 5 \cos^2 x + 7$, in the interval $0° \leq x \leq 360°$.

Giving each solution correct to 1 decimal place.

a)

$\tan^2 x - 2\tan x = 0$

$\tan x(\tan x - 2) = 0$

FACTORISE

TWO POSSIBLE SOLUTIONS

$\tan x = 0$

$\tan^{-1}(0) = 0$

$\tan x - 2 = 0$

$\tan x = 2$

$\tan^{-1}(2) = 63.43$

$63.4 + 180 = 243.4$

Tan REPEATS EVERY 180° SO ADD 180° TO FIND ALL SOLUTIONS

$x = 0, 180, 360$ AND $x = 63.4, 243.4$

b) i)

$11\sin x = 5\cos^2 x + 7$

$11\sin x = 5(1 - \sin^2 x) + 7$

$11\sin x = 5 - 5\sin^2 x + 7$

$5\sin^2 x + 11\sin x - 12 = 0$

USE $\cos^2 x = 1 - \sin^2 x$ TO GET EVERYTHING IN TERMS OF $\sin x$

EXPAND THEN REARRANGE

ii)

$5\sin^2 x + 11\sin x - 12 = 0$

$(5\sin x - 4)(\sin x + 3) = 0$

USING $11\sin x = 5\cos^2 x + 7$ WRITTEN IN THE FORM $5\sin^2 x + 11\sin x - 12 = 0$ FROM PART i)

FACTORISE USING $\sin x = s$ TO MAKE IT EASIER $(5s - 4)(s + 3)$

$5\sin x - 4 = 0$

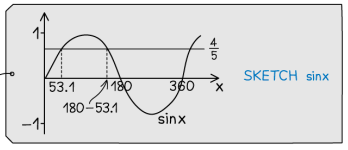
$5\sin x = 4$

$\sin x = \frac{4}{5}$

$\sin x + 3 = 0$

$\cancel{\sin x = -3}$

$\sin x = -3$ HAS NO SOLUTION



SKETCH $\sin x$

ONLY TWO VALID SOLUTIONS FOR GIVEN INTERVAL

$\sin^{-1}\left(\frac{4}{5}\right) = 53.1$

$180 - 53.1 = 126.9$

TWO SOLUTIONS $x = 53.1, 126.9$

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